

1. Microprocessor has a function of Fetching, Decoding and execution of instructions in a system.
3. Unidirectional buses are the lines or wires that carry signals in only one direction either from microprocessor to the memory or input/output device or memory or input device to the microprocessor. Example: Address buses are unidirectional buses that carry information in both directions. Example: Data buses.
2. (i) Microcontroller is the form of microprocessor designed for special purpose. Example: monitoring the temperature of the system.
2. (i) Microcontroller has limited number of functions as compared to microprocessor where microprocessor can be made flexible of performing more task than microcontroller.
- (ii) Microcontroller has few number of pins compared to microprocessor.
- (iii) Inside microcontroller there is microprocessor hence microprocessor is the basic function of microcontroller while microprocessor is independent in microcontroller.

4. Given:

STA 175H

STA = 4T

17H = 3T

53H = 3T

memory write = 3T

$4T + 3T + 3T + 3T = 13T$

$T = 2.5 \mu s$

$T_{\text{total}} = 13 \times 2.5 \mu s$

$T_{\text{total}} = 32.5 \mu s$

$\therefore$ Total time will be $32.5 \mu s$.

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i) immediate addressing, in this mode of addressing the operand is given in the instruction. example `MVI A, 23H`.

ii) memory direct addressing, in this mode the address of the memory location is given in the instruction. example `LDA 2300H`.

iii) Register direct addressing, in this mode the data transfer is between registers. example `MOV A, B`.

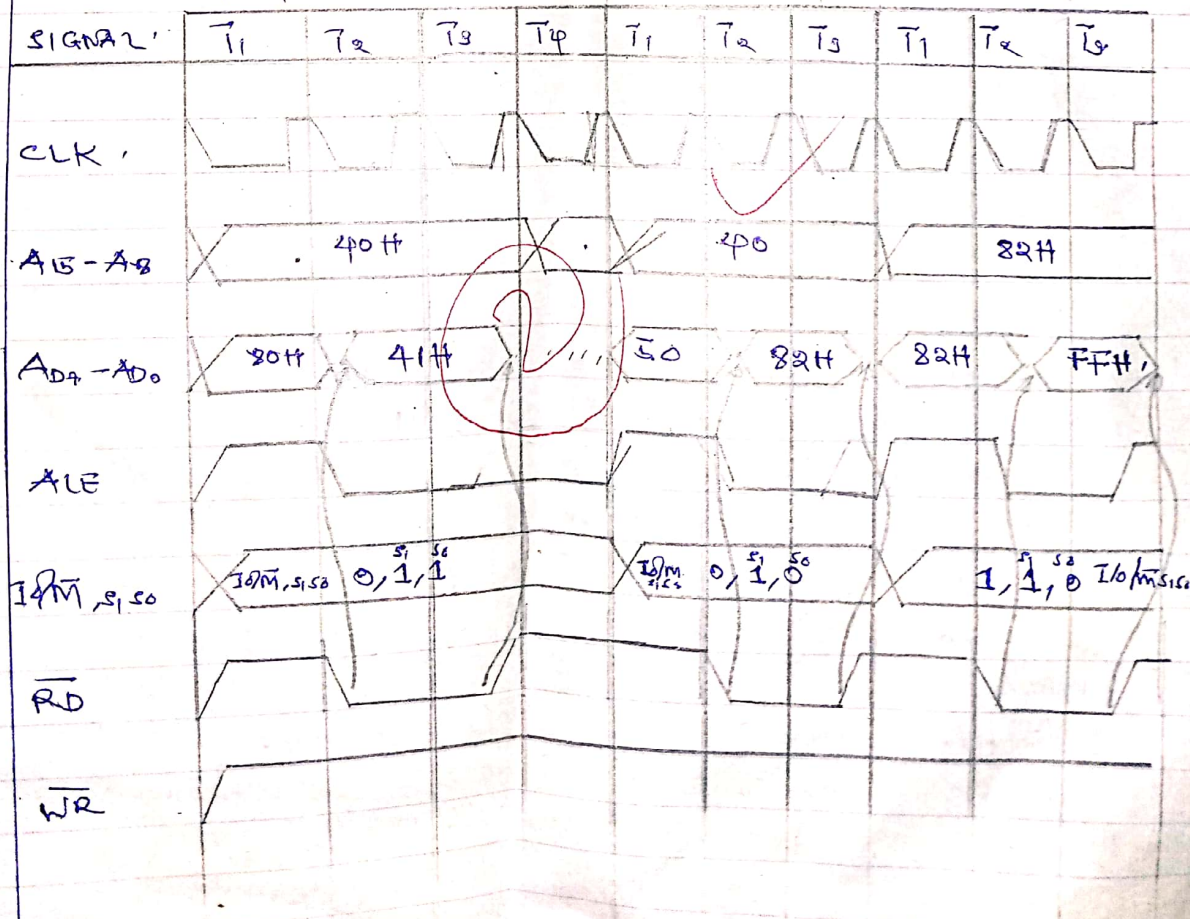
iv) indirect addressing, in this mode Register is used as a pointer to the memory location. example `LDAX B`.

v) implicit addressing, in this mode the data itself specifies the operand to be operated on. example `CMA`.

Q. Given: `IN 82H`. Assume the following

Address	Mnemonic	WZR	opcode
4080H	IN		41H
4058H			82H

- Assume also the value to be implied be FFH from 82H, opcode fetch and operand memory Read I/O Read,



7. ①

MVI A, 25H
MVI B, F8H
MVI C, 78H
MVI H, 48H
HLT

②

MOV L, B
~~HLT~~

③

STA 3004H
HLT

④

MOV M, C
~~HLT~~

⑤

CMA
~~HLT~~

⑥

HLT

PROGRAMME

7

① MVI A, 08H
LHLD 4004H
XCHG
LHLD 4002H
DAD D
JNE LOOP
INR A

LOOP: SHLD 4004H
STA 4006H

ADDITION PROCESS

9B23H = 10011011 00100011

4645H = 10000110 01000101

10011011 00100011
10000110 01000101

01000001 01100000
41 68

4004H	68H
4005H	41H
4006H	carry

DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING

MICROPROCESSOR EE 8312 TEST 1 May 8, 2023 time 1.5hrs

20 marks

Answer all questions.

1. What is the function of a microprocessor in a system?
2. Write three differences between microprocessor and microcontroller
3. With examples in 8085, differentiate between Unidirectional and Bidirectional system buses
4. How long would the processor take to execute the instruction STA 1753H if the T-state duration is $2.5\mu\text{s}$?
5. With examples explain the addressing modes in 8085 microprocessor
6. Draw and explain the timing diagram of the instruction IN 82H
7.
 - a) Develop an 8085 Microprocessor assembly program to perform the following:
 - i) Load 25H, F3H, 78H and 40H into registers A, B, C and H respectively
 - ii) Transfer the content of register B into register L
 - iii) Store the content of register A into memory location with address 3001H
 - iv) Store the content of register C into memory location pointed by HL register pair
 - v) Complement the content of register C
 - vi) Stop
 - b) Add the 16-bit number A645H in memory locations (4000H and 4001H) to the 16-bit number 9B23H in memory locations (4002H and 4003H). Store the result in memory locations 4004H and 4005H with the most significant byte in memory location 4005H

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